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MEDICAL INFORMATION PROCESSING DEVICE, MEDICAL INFORMATION PROCESSING METHOD, AND STORAGE MEDIUM

Abstract

A medical information processing device of an embodiment includes processing circuitry. The processing circuitry is configured to acquire a CT image of a chest of a subject, acquires an esophageal pressure and an airway pressure of the subject, generate a risk information map in a lung field of the subject on a basis of the CT image, the esophageal pressure, and the airway pressure, estimate risk information in the lung field of the subject on a basis of the risk information map, and display the risk information map and the risk information on a display.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
[0001] The present application claims priority based on Japanese Patent Application No. 2023-179486 filed Oct. 18, 2023, the content of which is incorporated herein by reference.
FIELD
[0002] Embodiments disclosed in the present specification and drawings relate to a medical information processing device, a medical information processing method, and a storage medium.
BACKGROUND
[0003] In the artificial respiration management of patients with acute respiratory distress syndrome (ARDS), a lung protective ventilation strategy based on limiting the tidal volume is recommended to prevent ventilator-associated lung injury due to alveolar hyperextension. Transpulmonary pressure, which is a determining factor of ventilator-associated lung injury, is measured by a transpulmonary pressure measurement method using an esophageal balloon or the like.
[0004] In the lung protective ventilation strategy in the related art, a uniform artificial respiration management is performed for all patients with the same tidal volume (6 ml/kg). As a result, the mortality rate of patients remains extremely high. For this reason, a method is required to establish an individualized lung protective ventilation strategy for each patient according to the degree of risk of ventilator-associated lung injury. In order to establish an individualized lung protective ventilation strategy, it is necessary to accurately ascertain the transpulmonary pressure, which is a determining factor of ventilator-associated lung injury that differs from patient to patient. However, transpulmonary pressure measurement methods using an esophageal balloon or the like in the related art reflect the transpulmonary pressure in a specific local lung region and are unable to ascertain the accurate transpulmonary pressure in the entire lung. As a result, it is not possible to appropriately evaluate the risk of ventilator-associated lung injury.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0005] FIG. **1** is a functional block diagram showing an example of a medical information processing device **100** according to an embodiment.
[0006] FIG. **2** is a flowchart showing an example of processing of the medical information processing device **100** according to an embodiment.
[0007] FIG. **3** is a diagram showing how a lung region and an esophageal pressure measurement position are determined in a CT image IMG**1** according to an embodiment.
[0008] FIG. **4** is a diagram showing how a transpulmonary pressure for each unit region is calculated in the CT image IMG**1** according to the embodiment.
[0009] FIG. **5** is a diagram showing an example of a transpulmonary pressure map according to an embodiment.
[0010] FIG. **6** is a diagram showing another example of a transpulmonary pressure map according

to an embodiment.
[0011] FIG. **7**A is a diagram showing an example of input and output of a risk estimation model M according to an embodiment.
[0012] FIG. **7**B is a diagram showing another example of input and output of the risk estimation model M according to an embodiment.

DETAILED DESCRIPTION

[0013] A medical information processing device, a medical information processing method, and a storage medium of an embodiment will be described below with reference to the drawings. The medical information processing device of the embodiment generates a risk information map showing a lung injury risk in the lung field of a patient on the basis of a CT image, esophageal pressure, and airway pressure of the subject (patient), estimates risk information in the lung field of the patient, and displays the risk information map and the risk information on a display, thereby enabling a detailed and accurate evaluation of the risk of lung injury.
[0014] A medical information processing device of an embodiment includes processing circuitry. The processing circuitry is configured to acquire a CT image of a chest of a subject, acquires an esophageal pressure and an airway pressure of the subject, generate a risk information map in a lung field of the subject on a basis of the CT image, the esophageal pressure, and the airway pressure, estimate risk information in the lung field of the subject on a basis of the risk information map, and display the risk information map and the risk information on a display.

<Configuration of Medical Information Processing Device **100**>

[0015] FIG. **1** is a functional block diagram showing an example of a medical information processing device **100** according to an embodiment. For example, the medical information processing device **100** is operated by a user such as a doctor or a technician who examines and treats patients. The medical information processing device **100** is, for example, a computer apparatus (a terminal device such as a personal computer or a tablet) used in a CT examination room, an operating room, a hospital room, and the like. The medical information processing device **100** is connected to external devices (medical apparatuses, modalities, and the like) such as an X-ray CT device D**1**, an artificial ventilator D**2**, and an esophageal pressure measuring device D**3** via a communication network NW such that they can communicate with each other. The medical information processing device **100** may be a server device that is connected to terminal devices used by doctors and others via a network.
[0016] The medical information processing device **100** includes, for example, a communication interface **110**, an input interface **120**, a display **130**, processing circuitry **140**, and a memory **150**. The communication interface **110** communicates with an external device via a communication network NW. The communication network NW may refer to a general information and communication network that uses electrical communication technology. For example, the communication network NW includes telephone communication line networks, optical fiber communication networks, cable communication networks, and satellite communication networks in addition to wireless/wired LANs such as a hospital backbone local area network (LAN) and the Internet. The communication interface **110** includes, for example, a network interface card (NIC) and an antenna for wireless communication.
[0017] The input interface **120** receives various input operations from a user (operator), converts the received input operations into electrical signals, and outputs the electrical signals to the processing circuitry **140**. For example, the input interface **120** includes a mouse, a keyboard, a trackball, a switch, a button, a joystick, a touch panel, and the like. The input interface **120** may be, for example, a user interface that receives voice input, such as a microphone. If the input interface **120** is a touch panel, the input interface **120** may also have the display function of the display **130**.
[0018] Note that in this specification, the input interface **120** is not limited to an interface that has physical operating parts such as a mouse and a keyboard. For example, examples of the input interface **120** also include electrical signal processing circuitry that receives an electrical signal

corresponding to an input operation from an external input apparatus provided separately from the device and outputs the electrical signal to control circuitry.

[0019] The display **130** provides various types of information to a user. The display **130** displays, for example, CT images and various measurement values acquired from external devices, a risk information map generated by the processing circuitry **140**, a graphical user interface (GUI) image for receiving various input operations from a user, and the like. The display **130** is, for example, a liquid crystal display or the like. The display **130** may be provided separately from the medical information processing device **100**. The display **130** is an example of a "display."

[0020] The processing circuitry **140** controls the overall operation of the medical information processing device **100**. The processing circuitry **140** includes, for example, an acquisition function **141**, a determination function **142**, a calculation function **143**, a generation function **144**, an estimation function **145**, and a display control function **146**. The processing circuitry **140** realizes these functions, for example, by a hardware processor executing a program stored in a storage device (storage circuitry).

[0021] The hardware processor refers to circuitry such as a central processing unit (CPU), a graphics processing unit (GPU), an application specific integrated circuit (ASIC), a programmable logic device (e.g., simple programmable logic device (SPLD) or complex programmable logic device (CPLD)), or a field programmable gate array (FPGA). Instead of being stored in the storage device, the program may be directly built into the circuit of the hardware processor. In this case, the hardware processor realizes the functions by reading and executing the program built into the circuit. The hardware processor is not limited to being configured as a single circuit and may be configured as a single hardware processor by combining a plurality of independent circuits to realize each function. The storage device may be a non-transitory (hardware) storage medium. Further, a plurality of components may be integrated into a single hardware processor to realize each function.

[0022] The acquisition function **141** acquires various types of medical information from external devices via the communication network NW. The acquisition function **141** includes, for example, an image acquisition function **141-1** and an internal pressure acquisition function **141-2**.

[0023] The image acquisition function **141-1** acquires CT images of the chest of a patient P from the X-ray CT device D**1**. The CT images include, for example, a plurality of tomographic images of the patient P sliced in the XY plane (A-A′ cross section in FIG. **1**) in the range of the chest (for example, at least two-thirds of the lower thoracic esophagus) in the body axis direction (Z direction) of the patient P placed on a bed device BD of the X-ray CT device D**1**. The body axis direction of the patient P is defined as the Z axis direction (the longitudinal direction of the tabletop of the bed device BD), the axis orthogonal to the Z axis direction and horizontal to the floor surface is defined as the X axis direction, and the direction orthogonal to the Z axis direction and perpendicular to the floor surface is defined as the Y axis direction. The image acquisition function **141-1** may acquire a CT image from a database (for example, a picture archiving and communication system (PACS)) in which captured CT images are stored. The image acquisition function **141-1** may acquire a single CT image of the patient P from the X-ray CT device D**1**. The image acquisition function **141-1** is an example of an "image acquirer."

[0024] The internal pressure acquisition function **141-2** acquires information indicating the airway pressure and esophageal pressure of the patient P. The airway pressure information is acquired, for example, from the artificial ventilator D**2** connected to the lungs via the mouth, throat, and the like of the patient P. The esophageal pressure information is acquired, for example, from the esophageal pressure measuring device D**3**. For example, an esophageal pressure tube or an esophageal pressure balloon kit for measuring an esophageal pressure is used as the esophageal pressure measuring device D**3**. For example, before or after CT imaging, an esophageal pressure balloon kit is placed in the esophagus of the patient P through the mouth or the like of the patient P. Thereafter, the esophageal balloon placed in the esophagus of the patient P is filled with an appropriate amount of air, the presence of a cardiac waveform and the appropriate placement position of the esophageal balloon are checked using the Baydur occlusion method or the like, and then the esophageal pressure is measured. The esophageal pressure measured in this manner is used as a substitute for the pleural pressure. The internal pressure acquisition function **141-2** is an example of an “internal pressure acquirer.”

[0025] The determination function **142** performs image analysis on an acquired CT image to determine the characteristics of the chest on the CT image. For example, any method such as segmentation using machine learning technology can be used as the image analysis. The determination function **142** includes, for example, a lung region determination function **142-1** and an esophageal pressure measurement position determination function **142-2**. The lung region determination function **142-1** determines a lung region indicating a region in which lungs are imaged in a CT image. The esophageal pressure measurement position determination function **142-2** determines an esophageal pressure measurement position indicating a position where an esophageal balloon is placed in the esophagus and the esophageal pressure is measured in a CT image. The determination function **142** may determine or correct a lung region and an esophageal pressure measurement position on the basis of information manually input by a user via the input interface **120**. In particular, ARDS patients often have bilateral infiltrative shadows in the dorsal part. Depending on the method of automatically determining a lung region, an infiltrative shadow may not be correctly recognized as being within a lung region. Therefore, the user can perform manual correction via the input interface **120**.

[0026] The calculation function **143** calculates a transpulmonary pressure for each unit region included in a region determined to be a lung region in a CT image. The calculation function **143** calculates the transpulmonary pressure for each unit region on the basis of a relative positional relationship of a unit region for which the transpulmonary pressure is to be calculated with respect to an esophageal pressure measurement position, a CT value of the unit region in the CT image, and actual measured values of the airway pressure and esophageal pressure. The unit region may correspond to each pixel in the CT image, or may be composed of a plurality of pixels. Details of processing of the calculation function **143** will be described later. The calculation function **143** is an example of a “calculator.” The calculation function **143** calculates the proportion of unit regions having transpulmonary pressures equal to or greater than a threshold value on the basis of the transpulmonary pressure for each unit region.

[0027] The generation function **144** generates a risk information map (transpulmonary pressure map, Lang stress map) showing a transpulmonary pressure for each unit region on the basis of a calculated transpulmonary pressure value for each unit region. The risk information map divides the lung region of the patient P into small regions (unit regions) and visualizes the amount, location, and pattern of risk regions for lung injury on the basis of transpulmonary pressure values. Details of processing of the generation function **144** will be described later. The generation function **144** is an example of a “generator.” That is, the generation function **144** generates a risk information map for the lung field of a subject on the basis of a CT image, an esophageal pressure, and an airway pressure.

[0028] The estimation function **145** estimates risk information for the lung region of the patient P on the basis of the risk information map. The risk information includes, for example, information indicating a high-risk region where the risk of lung injury is particularly high. The estimation function **145** estimates the risk information using, for example, a machine learning function using artificial intelligence (AI). The estimation function **145** may be realized by, for example, a deep neural network. Alternatively, the estimation function **145** may be implemented using other machine learning models such as a support vector machine, a decision tree, a random forest, and a logistic regression instead of a neural network. Details of processing of the estimation function **145** will be described later. The estimation function **145** is an example of an “estimator.” The estimation function **145** estimates risk information using a risk estimation model that has been trained to output information indicating risk information when information based on a risk information map or a combination of information based on a risk information map and subject information is input.

[0029] The display control function **146** displays, on the display **130**, a CT image and various measurement values acquired from external devices, a risk information map generated by the processing circuitry **140**, risk information, a GUI image for receiving various input operations from the user, and the like. The display control function **146** is an example of a "display controller." The display control function **146** superimposes the risk information map on the CT image and displays the same on the display. The display control function **146** displays the proportion of unit regions having transpulmonary pressures equal to or greater than a threshold value on the display.

[0030] The memory **150** is realized, for example, by a semiconductor memory element such as a random access memory (RAM) or a flash memory, a hard disk, an optical disc, and the like. The memory **150** stores, for example, a risk estimation model M used in estimation processing in the estimation function **145**. In addition, the memory **150** stores a CT image and various measurement values acquired from external devices, a risk information map generated by the processing circuitry **140**, and the like. Such data may be stored in an external memory with which the medical information processing device **100** can communicate, instead of (or in addition to) the memory **150**. The external memory is controlled by a cloud server that manages the external memory, for example, by the cloud server receiving a read/write request.

<Processing Flow>

[0031] Next, a flow of processing performed by the medical information processing device **100** will be described. FIG. **2** is a flowchart showing an example of processing performed by the medical information processing device **100** according to an embodiment. In the following, it is assumed that processing is started with the patient P placed on the bed device BD of the X-ray CT device D**1**. The processing of the flowchart shown in FIG. **2** is started, for example, when a user issues an instruction to start risk evaluation processing via the input interface **120**.

[0032] First, the image acquisition function **141-1** acquires CT images of the patient P from the X-ray CT device D**1** (step S**101**). The CT images include, for example, a plurality of tomographic images of the patient P sliced in the XY plane in the range of the chest in the body axis direction (Z direction) of the patient P placed on the bed device BD.

[0033] Next, the internal pressure acquisition function **141-2** acquires information indicating the airway pressure and esophageal pressure of the patient P (step S**103**). The airway pressure information is acquired from the artificial ventilator D**2**. The esophageal pressure information is acquired from the esophageal pressure measuring device D**3** using an esophageal balloon, for example. Note that the airway pressure measured with the same ventilator settings as those used during CT imaging may be acquired with respect to the inspiratory pressure and positive end-expiratory pressure (PEEP).

[0034] Next, the determination function **142** identifies one CT image of the slice position to be subjected to subsequent processing from among the acquired CT images (step S**105**). Next, the lung region determination function **142-1** performs image analysis on the extracted CT image to determine a lung region (step S**107**). Furthermore, the esophageal pressure measurement position determination function **142-2** performs image analysis on the extracted CT image to determine an esophageal pressure measurement position (step S**109**).

[0035] FIG. **3** is a diagram showing how a lung region and an esophageal pressure measurement position are determined in a CT image IMG**1** according to an embodiment. In the example of FIG. **3**, a lung region AR**1** corresponding to the right lung and a lung region AR**2** corresponding to the left lung are determined as lung regions, and further, an esophageal pressure measurement position ES indicating the position of the esophagus is determined as an esophageal pressure measurement position. If the slice position of the extracted CT image corresponds to the placement position of the esophageal balloon, the esophageal balloon will be located at the esophageal pressure measurement position ES.

[0036] Referring back to FIG. **2**, subsequently, the calculation function **143** calculates a transpulmonary pressure (also called a "local transpulmonary pressure") for each unit region in the regions determined to be lung regions in the extracted CT image (step S**111**). The calculation function **143** calculates the transpulmonary pressure for each unit region on the basis of a relative position of the unit region with respect to the esophageal pressure measurement position ES, a CT value of the unit region, and actual measured values of the airway pressure and esophageal pressure.

[0037] FIG. **4** is a diagram showing how a transpulmonary pressure for each unit region is calculated in the CT image IMG**1** according to an embodiment. As shown in FIG. **4**, small regions (unit regions PG) having a predetermined area are set on the lung region AR**1** and the lung region AR**2**. The unit region PG may correspond to one pixel of the CT image, or may be composed of a plurality of pixels. Here, for example, a case in which the position in the Y axis direction of the esophageal pressure measurement position ES is set as a height reference position Pb (height 0), and a transpulmonary pressure (local transpulmonary pressure) is calculated for a unit region PG**1** corresponding to a position h [cm] higher in the Y-axis direction than the height reference position Pb will be described. If the average CT value of the part corresponding to the height h with respect to the unit region PG**1** is H.sub.Avg, the airway pressure acquired in step S**103** is P.sub.Airway [cmH.sub.2O], and the esophageal pressure acquired in step S**103** is P.sub.Esophagus [cmH.sub.2O], the local transpulmonary pressure P.sub.Transpulmonary [cmH.sub.2O] in the unit region PG**1** is calculated by the following formula (1).

[00001] $P_{\mathrm{Transpulmonary}} = P_{\mathrm{Airway}} - [P_{\mathrm{Esophagus}} - h * \{1 - (H_{\mathrm{Avg}} / -1000)\}]$ Formula(1)

[0038] The average CT value H.sub.Avg of the part corresponding to height h is obtained by setting a rectangular region of interest ROI vertically downward from the unit region PG**1**, which is the calculation point of the local transpulmonary pressure, to the height reference position Pb, which is the height of the measurement position of the esophageal pressure, and averaging CT values of all pixels present in the region of interest ROI. Since a CT value is also a value indicating the density of a target region, information on the CT value can be used to calculate a transpulmonary pressure. The width of the region of interest ROI can be one pixel, or the region of interest ROI can have a width of a plurality of pixels having the unit region PG**1** as a center. This can reduce noise caused by local deviations (errors) in CT values.

[0039] When the calculation point of the local transpulmonary pressure is near the edge of the lung, as in the unit regions PG**2** and PG**3**, pixels outside the lung regions may be included in the region of interest ROI for calculating the average CT value. In this case, the average CT value H.sub.Avg may be obtained for pixels within the region of interest ROI and within the lung regions.

[0040] Further, when the calculation point of the local transpulmonary pressure is lower than the height reference position Pb, which is the measurement position of the esophageal pressure, as in the unit region PG**4**, the value of h in the above formula (1) is calculated as a negative value.

[0041] By repeating calculation of the local transpulmonary pressure with each unit region PG (each pixel or pixel group) within the lung regions as the calculation point of the local transpulmonary pressure, the numerical value of the transpulmonary pressure can be calculated for the entire lung region.

[0042] Referring back to FIG. **2**, the processing circuitry **140** determines whether calculation processing of transpulmonary pressures has been completed for all acquired CT images (step S**113**). If it is determined that calculation processing of transpulmonary pressures has not been completed for all CT images (step S**113**; NO), processing returns to step **105** and the same processing is performed on unprocessed CT images.

[0043] On the other hand, if it is determined that calculation processing of transpulmonary pressures for all CT images has been completed (step S**113**; YES), the generation function **144** generates a transpulmonary pressure map (risk information map) on the basis of the

transpulmonary pressure value for each unit region PG calculated for each CT image (step S**115**). Here, since a transpulmonary pressure map with respect to a plurality of tomographic images (CT images) in the range of the chest in the body axis direction of the patient P is obtained, a three-dimensional 3D transpulmonary pressure map is generated as a result. That is, the generation function **144** generates a three-dimensional risk information map in the body axis direction of the subject on the basis of CT images including a plurality of tomographic images of the chest captured in the body axis direction of the subject.
[0044] FIG. **5** is a diagram showing an example of a transpulmonary pressure map LSM**1** in which the numerical value of the transpulmonary pressure calculated for each unit region is superimposed on a CT image according to an embodiment. As shown in FIG. **5**, a user such as a doctor can ascertain the detailed and accurate risk of lung injury of the patient P by checking such a transpulmonary pressure map LSM**1**.
[0045] Furthermore, by preparing in advance a lookup table in which transpulmonary pressure values are associated with display colors or grayscales, the generation function **144** can generate the transpulmonary pressure map as a color map image or grayscale image. The height distribution of transpulmonary pressures in the lung regions and degree thereof can be intuitively ascertained as colors or grayscales. FIG. **6** is a diagram showing an example of a transpulmonary pressure map LSM**2** displayed as a grayscale image according to an embodiment. By checking the transpulmonary pressure map LSM**2** displayed as such a grayscale image (or color image), a user such as a doctor can intuitively and easily ascertain the risk state of lung injury of the patient P.
[0046] Referring back to FIG. **2**, subsequently, the estimation function **145** estimates risk information in the lung regions of the patient P on the basis of the transpulmonary pressure map (step S**117**). For example, the estimation function **145** estimates a high-risk region where the risk of lung injury is particularly high. The estimation function **145** estimates the risk information using, for example, a risk estimation model M pre-stored in the memory **150**.
[0047] FIG. **7**A and FIG. **7**B are diagrams showing input and output of the risk estimation model M according to an embodiment. The risk estimation model M shown in FIG. **7**A is a model trained to output risk information when a transpulmonary pressure map LSM (risk information map) or information based on the transpulmonary pressure map LSM (for example, numerical information on transpulmonary pressures, information on the proportion of the area of regions where transpulmonary pressures are equal to or greater than a threshold value to the area of the entire lung region, and the like) is input. The risk information includes, for example, information indicating a high-risk region and information indicating a prognosis prediction (a mortality rate, a treatment method, and the like). Further, the risk estimation model M shown in FIG. **7**B is a model trained to output risk information when a transpulmonary pressure map LSM (risk information map) or information based on the transpulmonary pressure map LSM (e.g., numerical information on transpulmonary pressures and the like) and patient information are input. The patient information includes, for example, information included in a chart (a medical history, a smoking status, and the like), clinical information, and the like. In such a risk estimation model M, risk information can be estimated in detail by determining a distribution pattern of the transpulmonary pressures in addition to the proportion of the area of regions where transpulmonary pressures are equal to or greater than the threshold value to the entire area of the lung regions. Even if the proportion of the area of the regions where the transpulmonary pressures are equal to or greater than the threshold value to the entire area of the lung regions is the same, a treatment method can be changed by ascertaining the risk level according to the distribution pattern. For example, when the transpulmonary pressure on the ventral side is higher than that on the dorsal side, the risk can be reduced by changing the position of a patient to a prone position. Further, there is a recent method of monitoring the state of the lungs in real time during artificial respiration management using Electrical Impedance Tomography (EIT). The transpulmonary pressure map LSM is initial values of the state of the lungs, and EIT is the state of the lungs at time t during artificial respiration management. The

transpulmonary pressure map LSM at time t may be estimated using a model that uses the transpulmonary pressure map LSM and EIT as inputs. In addition, the risk of developing ventilator-associated lung injury during artificial respiration management may be predicted using a model that uses the transpulmonary pressure map LSM and EIT as inputs.

[0048] Referring back to FIG. **2**, subsequently, the display control function **146** causes the display **130** to display the generated transpulmonary pressure map LSM and the estimated risk information (step S**119**). By checking the transpulmonary pressure map LSM and the risk information, the user such as a doctor can perform detailed and accurate risk evaluation of lung injury of the patient P. Accordingly, processing of this flowchart ends.

[0049] Although an example of a case in which the transpulmonary pressure is calculated for each unit region PG (each pixel) has been described above, grids may be formed with a predetermined size at the time of display, and a transpulmonary pressure map may be generated and displayed using a representative value such as the average value of transpulmonary pressures within each grid. A region where a transpulmonary pressure is equal to or greater than the threshold value may be highlighted and displayed. A region where a transpulmonary pressure is equal to or greater than the predetermined threshold value may be regarded as a region where the risk of ventilator-associated lung injury is high or a region where the risk of ventilator-associated complications is high (e.g., pneumothorax, pneumomediastinum, pneumatocele, and the like), and the region may be highlighted and displayed. For example, a display method such as surrounding the region on a CT image with a contour or overlaying the region may be adopted.

[0050] Further, a lung region may be divided into a plurality of sections, and statistics of transpulmonary pressures may be calculated for each section, and the statistics may be displayed or displayed as a color map image or a grayscale image according to a lookup table. The method of dividing a lung region into a plurality of sections is, for example, to divide the lung region into four sections by dividing the lung region into two sections, the left lung and the right lung, in the horizontal direction and dividing the lung region into two sections, the ventral side and the dorsal side, in the vertical direction. As statistics, the average, median, maximum, minimum, variance, or the like can be used. Such a display makes it possible to visualize a distribution of the risk of ventilator-associated lung injury. Furthermore, a bias in the risk level, such as a left-right ratio or a top-bottom ratio, may be quantified or verbalized and displayed.

[0051] As the aforementioned method of dividing into sections, division may be performed on the basis of a CT value range of a CT image. For example, the method includes a method of dividing a lung region into hyperinflated lungs of −1,000 to −900 HU, normal lungs of −900 to −500 HU, poorly aerated lungs of −500 to −200, and atelectasis of −200 to +100. A distribution range (maximum to minimum or the like) and a distribution pattern of transpulmonary pressures or h*{1−(H.sub.Avg/−1000)} may be displayed. Such a display allows the uniformity/non-uniformity of the air distribution in the lungs to be quantified and visualized. Further, an alert may also be issued to the user depending on the risk of ventilator-associated lung injury.

[0052] In addition, the transpulmonary pressure map LSM may be generated using values different from the airway pressure and esophageal pressure acquired in step S**103**. For example, if the transpulmonary pressure map LSM is generated using a value higher than the airway pressure acquired in step S**103**, a transpulmonary pressure map LSM can be obtained on the assumption that settings of the artificial ventilator are changed such that the airway pressure increases. By comparing with the transpulmonary pressure map LSM generated using the airway pressure acquired in step S**103**, it is possible to determine whether the settings of the artificial ventilator can be changed safely.

[0053] According to the medical information processing device **100** of the embodiment described above, it is possible to evaluate a lung injury risk in detail and accurately for the entire lungs in consideration of the lung condition of each individual patient. In addition, it is possible to perform artificial respiration management in accordance with the patient's condition, and it is possible to prevent worsening of lung injury by visualizing the risk early, and thus it is expected that the patient prognosis will improve.

[0054] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions, and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A medical information processing device comprising processing circuitry configured to: acquire a CT image of a chest of a subject; acquire an esophageal pressure and an airway pressure of the subject; generate a risk information map in a lung field of the subject on a basis of the CT image, the esophageal pressure, and the airway pressure; estimate risk information in the lung field of the subject on a basis of the risk information map; and display the risk information map and the risk information on a display.

2. The medical information processing device according to claim 1, wherein the processing circuitry is further configured to superimpose the risk information map on the CT image and display the CT image including the risk information map on the display.

3. The medical information processing device according to claim 1, wherein the processing circuitry is further configured to estimate the risk information using a risk estimation model trained to output the risk information when information based on the risk information map is input.

4. The medical information processing device according to claim 1, wherein the processing circuitry is further configured to estimate the risk information using a risk estimation model trained to output information indicating the risk information when information based on the risk information map and information on the subject are input.

5. The medical information processing device according to claim 1, wherein the processing circuitry is further configured to calculate a transpulmonary pressure for each unit region of the lung field in the CT image on a basis of CT values in the CT image, the esophageal pressure, and the airway pressure, and generate the risk information map indicating the transpulmonary pressure for each unit region.

6. The medical information processing device according to claim 5, wherein the processing circuitry is further configured to calculate the transpulmonary pressure for each unit region on a basis of a relative positional relationship between a measurement position of the esophageal pressure in the CT image and the unit region for which the transpulmonary pressure is to be calculated.

7. The medical information processing device of claim 5, wherein the processing circuitry is further configured to calculate a proportion of the unit regions having transpulmonary pressures equal to or greater than a threshold value on a basis of the calculated transpulmonary pressure for each unit region, and display the proportion of the unit regions having transpulmonary pressures equal to or greater than the threshold value on the display.

8. The medical information processing device of claim 1, wherein the processing circuitry is further configured to generate the three-dimensional risk information map in a body axis direction of the subject on a basis of the CT images including a plurality of tomographic images of the chest captured in the body axis direction of the subject.

9. A medical information processing method, using a computer, comprising: acquiring a CT image of a chest of a subject; acquiring an esophageal pressure and an airway pressure of the subject; generating a risk information map in a lung field of the subject on a basis of the CT image, the esophageal pressure, and the airway pressure; estimating risk information in the lung field of the subject on a basis of the risk information map; and displaying the risk information map and the risk information on a display.

10. A computer-readable non-transitory storage medium storing a program causing a computer to: acquire a CT image of a chest of a subject; acquire an esophageal pressure and an airway pressure of the subject; generate a risk information map in a lung field of the subject on a basis of the CT image, the esophageal pressure, and the airway pressure; estimate risk information in the lung field of the subject on a basis of the risk information map; and display the risk information map and the risk information on a display.